

SOLUTIONS

1. B is correct. The tenth percentile corresponds to the lowest 10 percent of the observations in the sample, which are in bins 1 and 2.
2. B is correct. The second quintile corresponds to the second 20 percent of observations. The first 20 percent consists of bins 1 through 4. The second 20 percent of observations consists of bins 5 through 8.
3. C is correct. A quartile consists of 25 percent of the data, and the last 25 percent of the 20 bins are 16 through 20.
4. B is correct. The center of the 20 bins is represented by the market capitalization of the highest value of the 10th bin and the lowest value of the 11th bin, which is 46.88.
5. B is correct. The interquartile range is the difference between the lowest value in the second quartile and the highest value in the third quartile. The lowest value of the second quartile is 34.72, and the highest value of the third quartile is 58.34. Therefore, the interquartile range is $58.34 - 34.72 = 23.62$.
6. B is correct. Quintiles divide a distribution into fifths, with the fourth quintile occurring at the point at which 80 percent of the observations lie below it. The fourth quintile is equivalent to the 80th percentile. To find the y th percentile (P_y), we first must determine its location. The formula for the location (L_y) of a y th percentile in an array with n entries sorted in ascending order is $L_y = (n + 1) \times (y/100)$. In this case, $n = 10$ and $y = 80\%$, so

$$L_{80} = (10 + 1) \times (80/100) = 11 \times 0.8 = 8.8.$$
 With the data arranged in ascending order (–40.33 percent, –5.02 percent, 9.57 percent, 10.02 percent, 12.34 percent, 15.25 percent, 16.54 percent, 20.65 percent, 27.37 percent, and 30.79 percent), the 8.8th position would be between the eighth and ninth entries, 20.65 percent and 27.37 percent, respectively. Using linear interpolation, $P_{80} = X_8 + (L_y - 8) \times (X_9 - X_8)$,

$$P_{80} = 20.65 + (8.8 - 8) \times (27.37 - 20.65)$$

$$= 20.65 + (0.8 \times 6.72) = 20.65 + 5.38$$

$$= 26.03 \text{ percent.}$$
7. B is correct. The fund's standard deviation of returns is calculated as follows:

Year	Return	Deviation from Mean	Deviation Squared
1	4.5%	0.0150	0.000225
2	6.0%	0.0300	0.000900
3	1.5%	–0.0150	0.000225
4	–2.0%	–0.0500	0.002500
5	0.0%	–0.0300	0.000900
6	4.5%	0.0150	0.000225
7	3.5%	0.0050	0.000025
8	2.5%	–0.0050	0.000025
9	5.5%	0.0250	0.000625